

Phase 1: Initial path generation

Input
Map M , NFZs, p_o , p_k

Run search algorithm
(A*, Dijkstra, or Theta*)

Path found?

No

No path found

Yes

Extract waypoints $\{p_1 \dots p_{k-1}\}$

Phase 2: QIEA refinement

Initialize QIEA population
 $P_{best} \leftarrow P_{init}$

Measure Q-bits $\rightarrow P_{refined}$

Evaluate fitness $F(P)$
 $w_1 \cdot \text{Length} + w_2 \cdot \text{WP} + \text{Penalty}$

$F(P_j) < F(P_{best})?$

Yes

$P_{best} \leftarrow P_j$

No

Update Q-bits via $R(\Delta\theta)$

Yes

$t \leq T_{max}?$

No

Output
 $P^* = P_{best} (L^*, W^*)$

